Where Identity Breaks: Occlusion-Anchored Evaluation of Tracking-by-Detection in Dense Indian Traffic

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Does the tracker that wins on MOTA and IDF1 also keep an auto-rickshaw's identity after it passes behind a bus?

Data	A 2-minute clip of a dense signalised Indian intersection from a fixed elevated viewpoint, annotated in CVAT (primary); MOT17 training sequences and KITTI tracking labels (harness validation and occlusion statistics); IDD-Temporal (optional spot check only)
Detector	YOLOv8 with COCO weights, run once and cached so that every tracker receives identical boxes; an oracle detector built from ground-truth boxes of visible objects only
Trackers	SORT at its default 1-frame lifetime and at ByteTrack's 30-frame lifetime (the primary comparison); ByteTrack; ByteTrack without its low-score second association (mechanism ablation)
Manipulation	Controlled detection dropout: a target's boxes are deleted for $k \in \{2, 5, 10, 20, 30, 60\}$ frames to create occlusions of known length
Deliverable	UCS532 sessional report in CVPR format: MOTA and IDF1 for ByteTrack versus SORT, and a case study of 10 identity switches with visual evidence and proposed fixes
Duration	7 weeks: Week 3 to Week 9, with the mandatory checkpoint at Week 6

1. Motivation and Central Idea

Tracking-by-detection performs well on the benchmarks it was designed around, and those benchmarks are pedestrian scenes. MOTChallenge was created to give multi-object tracking a common evaluation framework with shared data, shared detections and one evaluation tool [13]. On its MOT17 test set, ByteTrack reports 80.3 MOTA, 77.3 IDF1 and 63.1 HOTA, and 77.8 MOTA, 75.2 IDF1 and 61.3 HOTA on the more crowded MOT20 [21].

An Indian intersection differs in kind, not only in density. Two-wheelers, auto-rickshaws, cycles, pedestrians, buses and trucks share the road with weak lane discipline. Small road users pass behind large ones continuously, so occlusion is a persistent state of the scene rather than an occasional event. Indian datasets document markedly different object types and densities from those of datasets collected in developed cities [6, 15, 18]. The project brief states that Western benchmarks never contain these occlusion patterns. This proposal treats that statement as a hypothesis to be measured (O5), not as an assumption.

The problem with aggregate evaluation. MOTA and IDF1 are single numbers summed over a whole video. MOTA adds misses, false positives and identity switches; in dense scenes the misses dominate, so a tracker can raise MOTA by detecting more while losing identities at exactly the moments that matter. Luiten *et al*. argue that MOTA over-weights detection and IDF1 over-weights association [11]. Our analysis of the frozen MOTChallenge leaderboards (Sec. 4.3) supports this: MOTA correlates with detection accuracy at Spearman $\rho = 0.95$–0.99, but with association accuracy at only 0.76–0.93. Neither aggregate number says *where* an identity was lost, or *why*.

The detector confound. Every "ByteTrack versus SORT" number also compares two detectors unless the detections are held fixed. COCO has no auto-rickshaw class, and detectors trained on Indian CCTV imagery report up to 31.5% higher mAP than COCO-pretrained baselines [15]. The leaderboards show the same effect (Tab. 3, finding 4). If YOLOv8 misses an occluded auto-rickshaw for ten frames, the identity switch that follows is a detector failure,

not a tracker failure.

Related work. Tracking-by-detection grew from SORT [3], a Kalman filter with IoU matching, through the appearance-based DeepSORT [20] and StrongSORT [7], to ByteTrack [21], which recovers low-score boxes, and BoT-SORT [1] and OC-SORT [4]; OC-SORT targets occlusion and non-linear motion directly. Benchmarks moved the same way: MOT20 for crowds [5], DanceTrack for look-alike objects with non-linear motion [17], and KITTI [8], UA-DETRAC [19] and TrafficMOT [10] for vehicles. All rank trackers by aggregate scores. Indian work has concentrated on detection datasets [15, 18], and tracking systems built for Indian roads, such as the YOLOv8 and DeepSORT counter HETRA [16], report detection and counting accuracy but no identity metrics. We found no study of where identities are lost in dense Indian traffic, which is the gap addressed here.

Central idea. The unit of evaluation should be the occlusion event, not the frame. Every occlusion in the annotated clip is recorded (which object was hidden, by what, and for how long), and each tracker is asked one question per event: does the object leave the occlusion with the identity it entered with? Two controlled manipulations follow. Feeding the trackers ground-truth boxes removes the detector, so any remaining switch is an association failure. Deleting a target's boxes for a chosen number of frames creates an occlusion of known length, which yields a curve of identity survival against occlusion duration. Because that curve is caused by our manipulation, it answers a practical question directly: how long can an object remain hidden before each tracker forgets it?

2. Objectives

Primary objectives.

O1 Report MOTA, IDF1 and HOTA for ByteTrack and SORT on the annotated clip using identical cached YOLOv8 detections, and determine whether the same ranking holds when trackers are scored per occlusion event. If it holds, the aggregate metrics are an adequate summary here and we report that; if it does not, the aggregate hides the failure that the brief is about.

O2 Locate and attribute every identity switch: inside or outside an occlusion event, and caused by a detector miss, an association error, motion-model drift or an annotation error. This produces the required 10-switch case study, selected by rule (Sec. 5.5).

O3 Measure, per tracker, how identity survival falls as a controlled occlusion lengthens, and report the occlusion ceiling k_{50}: the gap length at which half of the identities are lost.

Secondary objectives.

O4 Test ByteTrack's own explanation of its gain, namely that low-score boxes are often occluded objects whose association recovers identities [21], by switching the second association off and by replacing the detector with ground truth.

O5 Test the brief's premise by measuring occlusion statistics (events per object per minute, median occlusion length, share of objects ever occluded) with one definition on the Indian clip, on MOT17 (Western pedestrians, including static elevated views) and on KITTI (Western vehicles, dashcam view). Each comparator shares one property with our clip, so a difference found against both cannot be explained by viewpoint or object type alone.

O6 Propose a fix for each case-study switch and test the two or three cheapest fixes on the same events, reporting new failures alongside removed ones.

3. Novelty

The novelty is not a new tracker but an evaluation design that turns one short, self-annotated clip into a controlled experiment on where and why identity is lost.

N1. The occlusion event is the unit of evaluation. Each occlusion becomes a record (occludee and occluder class, duration, maximum hidden fraction) with one outcome per tracker. Both trackers face identical events, so arms are compared by their paired difference in survival, with intervals that resample whole tracks (Sec. 6.3).

N2. Occlusion length is a swept axis. Deleting a visible target's detections for k frames gives a survival curve per tracker, in which the only difference between $k = 5$ and $k = 30$ is the gap we introduced.

N3. Detector and tracker are separated. Trackers run on cached YOLOv8 boxes and on an oracle that supplies a ground-truth box only while the object is visible (Sec. 5.2). A switch that survives the oracle run is an association failure.

N4. The mechanism is tested, not assumed. ByteTrack's gain should concentrate in partial occlusions, shrink without the second association, and vanish on oracle boxes, which have no low-score tier. If it appears elsewhere, ByteTrack wins here for another reason.

N5. Fairness is controlled explicitly. SORT's code deletes a track after more than one missed frame (`max_age` = 1), while ByteTrack keeps it for 30 frames at 30 fps (`track_buffer` = 30) [3, 21]. Under defaults, any occlusion of two or more frames ends a SORT identity

Table 1. Why each objective is not already answered in the literature.

	Why it is not already answered
O1	Tracker papers report aggregate MOTA, IDF1 and HOTA per benchmark [4, 21]. Per-event identity survival is rarely reported, and we have not found it reported for heterogeneous traffic.
O2	Identity switches are counted (IDSW), not explained. A count cannot distinguish a tracker that fails at occlusions from one that fails at crossings.
O3	Trackers are compared at a single setting of their track-lifetime parameter. How survival decays with occlusion length is not normally plotted.
O4	ByteTrack's ablations use pedestrian benchmarks and its own detector [21]. Whether the mechanism transfers to two-wheelers hidden behind buses is untested.
O5	Indian datasets argue that their scenes are denser and more heterogeneous, but report detection statistics, not occlusion statistics under a tracking definition [6, 15, 18].
O6	Fixes for identity switches usually take the form of new trackers. Small, targeted fixes tested on the exact failures that motivated them are rarely reported.

Table 2. Data inventory.

Source	Contents	Role
Own clip	2 min, fixed elevated camera, ∼3,600 frames at 30 fps; CVAT tracks with an occluded flag per box	Primary evaluation set; source of all occlusion events
MOT17 train [12]	Pedestrian sequences with public ground truth, including per-box visibility; CC BY-NC-SA 3.0	Harness validation and occlusion statistics (O5)
KITTI labels [8]	Tracking training labels only (9 MB): cars and pedestrians from a dashcam	Western vehicle comparator for O5
IDD-Temporal [9]	∼30 frames per snippet, dashcam, no tracking IDs	Optional spot check on a moving camera
UVH-26 [15]	26,646 Bengaluru CCTV images, 1.8M boxes, 14 vehicle classes, released detector weights	Stretch arm: Indian-domain detector

by construction, so the primary comparison matches track lifetime.

4. Data

4.1. Verified sources

The brief offers a self-annotated clip or IDD-Temporal frames. IDD-Temporal provides the ± 15 frames around images of the IDD segmentation and detection sets, so each snippet holds about 30 frames from a moving vehicle [9]. The release describes frames only, and we found no tracking-identity annotations. It cannot contain an occlusion longer than about one second and would require our own annotation, so the self-recorded clip is the primary source (Tab. 2).

4.2. Annotation plan

The clip is annotated in CVAT track mode: boxes are drawn on keyframes, interpolated, and corrected by hand around every occlusion. Classes are coarse (pedestrian, two-wheeler, three-wheeler, car, bus or truck, cycle), and an evaluation region excludes the far background where road users are only a few pixels tall. Export uses the MOT format so that TrackEval reads it directly.

Annotation load. Annotation is the largest risk, so it is estimated, then measured. With N objects visible per frame and a keyframe every 10 frames, the clip needs about $3{,}600N/10$ keyframe boxes plus corrections around occlusions. At an assumed 5 s per box, $N = 40$ gives about 20 hours. These are assumptions: on the first day of Week 3 a timed 5-second pilot (150 frames) is extrapolated to the whole clip. If the projection exceeds a 25-hour budget, the clip is cut to 60 seconds or the evaluation region is narrowed before annotation continues, and the decision is recorded.

Pre-annotation is a known trap. If YOLOv8 boxes seed the ground truth, objects that YOLOv8 misses tend to be missing from the ground truth as well, and YOLOv8-based trackers appear better than they are. Ground truth is therefore drawn manually, or seeded by a different detector and audited frame by frame for missing objects.

Constraints that shape the design. *One clip is a small sample.* No claim is made about Indian traffic in general; the statistical unit is the occlusion event (Gate 1 requires at least 50) and the injected gap. *One annotator is a single point of failure.* A 10-second segment is re-annotated blind after at least one week, and identity agreement between the passes is reported. The gap sweep does not depend on clip length, so the core results survive the 60-second fallback.

Privacy. The clip shows faces and number plates on a public road. Raw video remains private, and every figure in the report blurs faces and plates.

4.3. Evidence from the frozen MOTChallenge leaderboards

MOTChallenge's evaluation server is closed; its results pages are a snapshot of 16 April 2026 with per-sequence metrics for every entry [13]. We analysed the MOT16, MOT17 and MOT20 tables (169, 218 and 116 trackers with HOTA); Tab. 3 lists five findings. Entries differ in detectors, training data and tuning, so these are patterns, not causes. Densities use sequence frame counts from the benchmark papers [5, 12], which are not in the result files and were checked against those papers.

5. Methods

5.1. What the system outputs

The unit of evaluation is the occlusion event: a record of (occludee class, occluder class, duration, maximum hidden fraction, detector, tracker) with one outcome (kept, switched or lost), plus the image strip around it for visual audit. The same tracker outputs are also scored by MOTA, IDF1 and HOTA with TrackEval, so the aggregate and event-level rankings come from identical runs.

5.2. Arms to be compared

All arms read the same cached detections (Tab. 4). Each arm runs on two inputs. The first is YOLOv8 with COCO weights (vehicle and person classes, tracked class-agnostically because COCO has no auto-rickshaw class), run once at a confidence threshold of 0.1, below the Ultralytics default, so that ByteTrack's low-score tier (scores 0.1–0.5) exists in the cache; SORT receives the boxes scoring at least 0.5, the same set as ByteTrack's first round. The second is an oracle built from the ground truth. Because CVAT keeps interpolated boxes through occlusions, a direct copy would never hide anything, so the oracle drops a box whenever the object is marked occluded or outside, or its hidden fraction (Eq. (1)) is at least 0.5. Oracle boxes carry score 1.0, so ByteTrack's low-score round is inactive on this input. OC-SORT [4] and the UVH-26 detector [15] are stretch arms, added only if Gate 3 passes with time remaining.

5.3. Occlusion events

In an elevated view, the box whose lower edge is lower in the image is nearer the camera. For object i at frame t, the hidden fraction is the share of its box $B_i(t)$ covered by nearer boxes:

$$\mathrm{occ}_i(t) = \frac{\left|B_i(t) \cap \bigcup_{j \in F(i,t)} B_j(t)\right|}{|B_i(t)|}, \tag{1}$$

where $F(i,t)$ is the set of objects in front of i. Box overlap overstates true occlusion, because a box also covers background around the object; the Gate 1 spot checks therefore report how often Eq. (1) and the annotator's occluded flag disagree. An occlusion event is a maximal run of frames with $\mathrm{occ}_i(t) \geq 0.5$, or with the object marked occluded or outside, preceded and followed by visible frames. Tracker boxes are matched to ground-truth objects in each frame by the Hungarian algorithm at IoU ≥ 0.5, the threshold used for MOTA. The entry identity is the tracker ID matched to the object in the last visible frame before the event. The outcome is *kept* if the same ID is matched within $w = 5$ frames after the event, *switched* if a different ID is matched, and *lost* otherwise. The identity survival rate over a set of events E is

$$S = \frac{|\{e \in E : e \text{ kept}\}|}{|E|}. \tag{2}$$

For O5 the same definition, Eq. (1) included, is applied to the ground-truth boxes of MOT17 and of the KITTI tracking training labels, a 9 MB download that needs no images; MOT17's annotator visibility is reported only as a secondary check.

5.4. The controlled gap sweep

Gap targets are drawn from the ground truth with a fixed seed, stratified by class: up to 10 each of pedestrians, two-wheelers, three-wheelers, cars and buses or trucks, 50 in total where available, with the realised composition reported. A target needs only to be clearly visible ($\mathrm{occ} < 0.1$) for 10 frames before and 10 frames after the gap; during the gap its boxes are deleted anyway. Requiring a long, fully visible run would favour large, slow vehicles over two-wheelers weaving through traffic. For each target and each $k \in \{2, 5, 10, 20, 30, 60\}$ frames (about 0.07–2 s at 30 fps), the target's boxes are deleted from the detection file for k frames and every arm is re-run, giving $S(k)$ per arm under the rule above. The occlusion ceiling k_{50} is found by linear interpolation between the grid points $k_a < k_b$ where S crosses 0.5:

$$k_{50} = k_a + \frac{S(k_a) - 0.5}{S(k_a) - S(k_b)}\,(k_b - k_a), \tag{3}$$

and is reported as "> 60" if S never falls below 0.5.

Under public defaults the outcome is largely fixed by configuration: SORT deletes a track after more than one missed frame and ByteTrack after more than 30, so the default curves mostly restate those settings. The sweep is therefore led by matched-lifetime runs in which both trackers keep lost tracks for 90 frames, longer than the largest gap. There, $S(k)$ measures what remains: whether the Kalman prediction carried through the gap still lies close enough to the reappearing object to be matched. The default curves are reported only as a check that each configuration behaves as documented.

The sweep buys an occlusion of exactly known length whose only effect is the missing boxes. It costs realism:

Table 3. Five findings from the frozen MOTChallenge leaderboards (snapshot of 16 April 2026), and what each changes in this proposal. ρ is Spearman rank correlation over overall results.

Finding	Evidence	Consequence here
1. Metrics agree on weak versus strong trackers, but not among strong ones	$\rho(\text{MOTA}, \text{IDF1}) = 0.91$–$0.96$ overall, but only 9, 4 and 2 of the top 10 by MOTA are in IDF1's top 10 (MOT16, 17, 20)	O1 matters most among strong trackers, the regime of this study
2. MOTA is mostly a detection score	$\rho(\text{MOTA}, \text{DetA}) = 0.95$–$0.99$, but $\rho(\text{MOTA}, \text{AssA}) = 0.76$–$0.93$; the gap is largest on MOT20 with private detections (0.98 vs. 0.76)	MOTA is reported as required, never as evidence about identity
3. ByteTrack's lead is in detection, not association	Among 115 MOT17 private-detection trackers, ByteTrack ranks 34th by MOTA but 53rd by AssA. OC-SORT has lower MOTA (78.0 vs. 80.3) but higher AssA (63.4 vs. 62.0)	Motivates the second-round ablation (N4)
4. The detector moves the numbers more than the tracker	OC-SORT scores 58.2 MOTA with public detections and 78.0 with its own; SORT17 (43.1) and ByteTrack (80.3) used different detections	Motivates fixed detections and the oracle (N3)
5. Crowd density alone does not predict identity loss	The densest sequences (MOT20-04, MOT17-03; about 132 and 70 people per frame) have the highest AssA; over 7 static-camera sequences $\rho(\text{density}, \text{AssA}) = +0.25$	Score occlusion events, not density (O5); 7 sequences give a hint only

Table 4. Tracker arms. All arms receive identical detections.

	Arm	Purpose
a	SORT, defaults (1-frame lifetime) [3]	Brief's baseline: Kalman filter with IoU matching of high-score boxes
a′	SORT, lifetime matched to b	Separates better association from longer track retention (N5)
b	ByteTrack, defaults (30-frame lifetime) [21]	Brief's main method: second association round for low-score boxes
c	ByteTrack, second round off	Mechanism ablation (O4); identical to b otherwise

Table 5. Ablation arms and the question each answers.

	Comparison	Question
A	SORT vs. ByteTrack on YOLOv8 by MOTA, IDF1, HOTA	The brief's required numbers
B	Same runs scored per event (S, paired difference)	Does the aggregate ranking survive at occlusions? (O1)
C	Same trackers on oracle boxes	How much of each switch count is the detector's? (N3)
D	ByteTrack with and without the second round	Is the gain where ByteTrack says it is? (O4)
E	$S(k)$ per arm at matched lifetime; defaults as a check	How long can an object stay hidden? (O3)
F	Ten switches plus cheapest fixes	Which failures are fixable, and at what cost? (O2, O6)

a real occluded object first produces partial, low-score and shifted boxes, which is precisely what ByteTrack's second round uses. The sweep therefore isolates the motion model and track lifetime, while natural events remain the test of the low-score mechanism. The two are always reported separately.

5.5. Choosing the 10 case-study switches

The case study is selected by rule so that it is not a gallery of the most dramatic failures. All switches of ByteTrack on YOLOv8 boxes are labelled by cause (detector miss, association error, motion drift, annotation error) and by occludee class. Ten are drawn with a fixed seed, stratified so that every cause with at least two switches is represented. Each case receives a strip of frames before, during and after the switch, the detection-score trace, the Kalman-predicted box, the diagnosed cause and a proposed fix.

5.6. Ablation design

Table 5 lists the experimental arms. Arm B decides whether the study has a finding beyond the brief; if the rankings agree everywhere, that agreement is itself reported.

6. Evaluation Protocol

6.1. Metrics

All aggregate metrics are computed with the official TrackEval implementation and are never re-implemented. Its MOTChallenge evaluator scores one class (pedestrian, id 1) and discards other ground-truth classes, so every row is written as class 1 under a custom benchmark and the real classes are kept separately for stratification. The perfect-tracker control (Sec. 6.5) would expose any mistake here. With per-frame false negatives FN_t, false positives FP_t, identity switches IDSW_t and ground-truth boxes GT_t,

$$\text{MOTA} = 1 - \frac{\sum_t (\text{FN}_t + \text{FP}_t + \text{IDSW}_t)}{\sum_t \text{GT}_t} \tag{4}$$

[2]. With identity-level true positives, false positives and false negatives under a one-to-one trajectory matching [14],

$$\mathrm{IDF1} = \frac{2\,\mathrm{IDTP}}{2\,\mathrm{IDTP} + \mathrm{IDFP} + \mathrm{IDFN}}. \tag{5}$$

HOTA is the geometric mean of detection accuracy and association accuracy at localisation threshold α, averaged over α [11]:

$$\mathrm{HOTA}_\alpha = \sqrt{\mathrm{DetA}_\alpha \cdot \mathrm{AssA}_\alpha}. \tag{6}$$

MOTA and IDF1 are reported because the brief requires them; HOTA balances detection and association; IDSW is the raw count the case study explains; the event-level survival S (Eq. (2)) and the gap-sweep curve $S(k)$ with its ceiling k_{50} are the primary measures.

6.2. Pre-declared expected results

These are expectations to be tested, not results. ByteTrack should outperform SORT on MOTA and IDF1 overall. Against the matched-lifetime SORT (a′), we expect ByteTrack's advantage at natural events to concentrate in partial occlusions ($0.5 \leq \mathrm{occ} < 0.9$), where low-score boxes still exist, and to largely disappear in two cases: full occlusions longer than the track lifetime, and the oracle run, which has no low-score tier. The leaderboard evidence (Tab. 3, finding 3) makes a smaller-than-expected association gain plausible. In the matched-lifetime sweep we expect survival to fall with k for both trackers, since a constant-velocity prediction drifts; whether ByteTrack's curve lies above SORT's is open.

6.3. Statistical analysis

Events are not independent: one object may be occluded several times, and one bus may hide several two-wheelers at once. Resampling individual events would give intervals that are too narrow. Confidence intervals (95%) therefore come from a cluster bootstrap that resamples occludee tracks, not events (10,000 resamples), and the primary comparison between two arms is their difference in S over the same events, with its cluster-bootstrap interval. McNemar's test, with b events kept only by arm 1 and c kept only by arm 2,

$$\chi^2 = \frac{(|b-c|-1)^2}{b+c}, \tag{7}$$

is reported only as a descriptive secondary statistic, because it assumes independent pairs. With 50 events and survival near 0.7, a single arm's 95% interval is already about ± 0.13 before clustering widens it, so differences in S below about 0.15 may not be resolved. Gate 1 therefore targets 100 events and treats 50 as the floor, and unresolved differences are reported as such. The pre-registered criterion is stated for the matched comparison (a′ against b), because under defaults SORT loses every identity hidden for two or more frames and the criterion would pass automatically: at least half of ByteTrack's reduction in switches relative to a′ must fall inside occlusion events. If the reduction is spread evenly instead, ByteTrack is winning on this footage for a reason other than the one it states, and we report that.

6.4. Tuning and held-out data

No model is trained, so no training split can leak. The remaining risk is tuning thresholds on the reported data. Detector and tracker thresholds stay at public defaults. The only tuned values, the occlusion threshold and the post-event window w, are fixed on the first 20 seconds of the clip (10 seconds if the clip is cut to 60), which are then excluded from every reported number. Gap-sweep targets are drawn once with a fixed seed and reused for every arm.

6.5. Mandatory negative controls

1. **Perfect tracker.** The ground truth is evaluated as a tracker output. MOTA, IDF1 and HOTA must equal 100 and IDSW must equal 0; otherwise the format, frame indexing or class mapping is wrong.
2. **Single swap.** In a copy of the ground truth used as tracker output, two objects exchange IDs from one frame onward. IDSW must equal exactly 2, and only identity metrics may fall.
3. **Zero gap.** The gap sweep at $k = 0$ must give $S = 1.0$ for every arm; anything lower indicates a fault in the deletion script.
4. **Matched lifetime.** Arm a′ appears in every table, so that "ByteTrack is better" never reduces to "ByteTrack waits longer".
5. **Annotator agreement.** The blind re-annotation of 10 seconds is reported beside every switch count.

6.6. Mandatory stratified reporting

Every headline result is also reported by occludee class, by occluder class, by occlusion-length bin, and by detector input (YOLOv8 or oracle). A result driven by a single class pair is stated as such and is not averaged away.

7. Work Plan and Gates

Each phase ends with a pass-or-fail gate, and the next phase starts only when it passes (Fig. 1): a mistake found in Week 4 costs a day, but in Week 9 it costs the project.

Phase 0: audit (Week 2, current). ✓ MOTChallenge metrics and their biases reviewed. ✓ IDD-Temporal checked and demoted to a spot check. ✓ ByteTrack, SORT and OC-SORT code and UVH-26 weights located. ✓ Frozen leaderboards analysed (Sec. 4.3). ☐ Intersection chosen, camera position tested and filming permission confirmed where required.

Phase 1: clip, annotation and harness (Weeks 3–4). *Gate 1:* annotation pilot timed and the full-clip projection recorded; clip recorded and annotated in MOT format; perfect-tracker and single-swap controls pass; at least 50 occlusion events extracted (target 100), 10 checked by eye; blind re-annotation completed.

Phase 2: detections and trackers (Weeks 4–6). *Gate 2, the Week 6 checkpoint:* YOLOv8 detections cached with model version and thresholds recorded; arms a, a′, b and c run end to end on both inputs; MOTA, IDF1, HOTA and IDSW table complete (arms A and C); track lifetimes set as documented and logged with every run; repository link submitted with the table and the control outputs.

Phase 3: primary experiment (Weeks 6–8). *Gate 3:* S with cluster-bootstrap intervals and paired differences for every arm (B); second-round ablation (D); gap sweep with the zero-gap control, realised target composition, and k_{50} at matched lifetime (E); every claimed switch saved as an image strip. Stretch arms are added only after this gate passes.

Phase 4: case study and report (Weeks 8–9). *Gate 4:* ten switches selected by rule with strips, score traces, causes and fixes; cheapest fixes re-run (F); premise statistics for the clip, MOT17 and KITTI (O5); stratified results; a single script regenerates every table and figure from a clean checkout; limitations written before figures are polished; AI-assistance declaration completed.

8. Planned Figures

Six figures from the Phase 3 outputs: (1) an annotated frame and the pipeline; (2) arms ranked by MOTA and IDF1 against S, where crossing lines are the result; (3) $S(k)$ at matched lifetime with k_{50}; (4) switches avoided by ByteTrack per occlusion bin, with and without the second round; (5) switch counts on YOLOv8 and oracle boxes by cause; (6) the ten case-study switches as blurred frame strips.

9. Timeline

If the project falls behind, work is dropped in this order: the stretch arms; the IDD-Temporal spot check; the $k = 60$ column of the gap sweep; the fix re-runs of O6, keeping the proposed fixes as text. Phases 1–3 and objectives O1–O3 are not droppable.

Compute. No training is required. YOLOv8 inference on about 3,600 frames runs once on a laptop GPU (Apple Silicon) or a free Colab session and is cached. SORT and ByteTrack on cached boxes run on a CPU, so the full gap sweep (6 tracker settings $\times$ 2 inputs $\times$ 6 gap lengths $\times$ 50 targets, about 3,600 short runs) is inexpensive. Every script writes its outputs per run, so a lost session costs minutes.

10. Risk Register and Limitations

Table 6 lists the main risks and responses. The following limitations are stated in advance.

- One intersection, one viewpoint and one time of day. Nothing here generalises to night, rain, dashcam footage or other cities, and no such claim is made.
- One annotator. Agreement is measured on a 10-second sample, which does not replace several independent annotators.
- Intervals resample occludee tracks but not occluders, so dependence through a shared occluder is not modelled.
- The "in front" rule in Eq. (1) uses image position, not depth, and box overlap overstates pixel occlusion. Both suit an elevated camera better than street level, and the disagreement with the annotator's flags is measured at Gate 1.
- The core study contains two motion-only trackers. Appearance-based trackers such as DeepSORT [20] may behave differently under long occlusions, and this study makes no claim about them.

11. Expected Contributions

C1. MOTA, IDF1 and HOTA for ByteTrack and SORT on dense Indian intersection footage, with the same runs scored per occlusion event.

C2. An occlusion ceiling per tracker at matched track lifetime, reflecting motion prediction rather than configuration.

C3. A test of ByteTrack's stated mechanism on heterogeneous traffic, with detector and tracker errors separated.

C4. A rule-selected case study of ten identity switches with causes, proposed fixes and tested cheap fixes.

C5. A reproducible harness for any annotated traffic clip, and occlusion statistics comparing it with MOT17 and KITTI.

12. References and Tooling

All references were checked against their arXiv, publisher or project pages on 11 October 2026, and the tracker defaults quoted here against the public source code. Code and tools: TrackEval (github.com/JonathonLuiten/TrackEval); CVAT (github.com/cvat-ai/cvat); Ultralytics YOLOv8 (github.com/ultralytics/ultralytics); ByteTrack (github.com/ifzhang/ByteTrack); SORT (github.com/abewley/sort); OC-SORT (github.com/noahcao/OC_SORT); UVH-26 (huggingface.co/datasets/iisc-aim/UVH-26). The leaderboard analysis script is in the project repository.

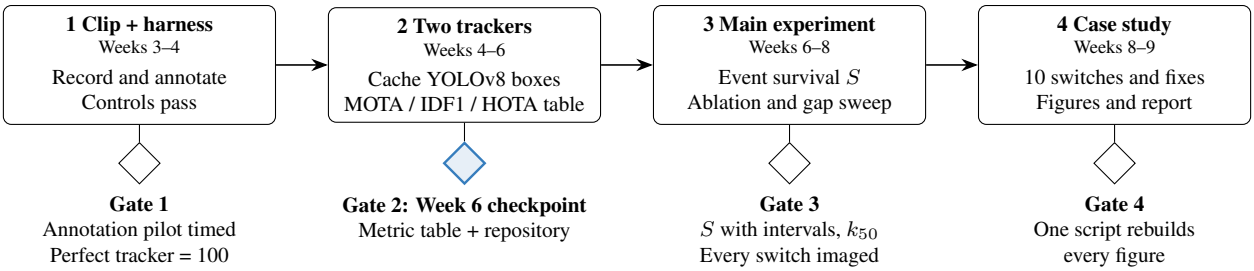


Figure 1. Work plan. Each phase starts only after the previous gate passes; Gate 2 is the mandatory Week 6 checkpoint. Phase 0 (audit) is completed in Week 2.

Table 6. Risk register.

Risk	Response
Perfect-tracker control does not return 100	Stop. This is a format, frame-index or class-mapping error that silently invalidates every later number; Gate 1 catches it in Week 3.
The Week 3 pilot projects more than 25 hours of annotation	Cut to 60 seconds or narrow the evaluation region before continuing, and state this. The gap sweep does not depend on clip length, so O3 and O4 survive intact.
Fewer than 50 natural occlusion events	Extend the clip if annotation time allows; otherwise report natural-event results as descriptive and rest quantitative claims on the gap sweep.
YOLOv8 misses so much that tracking metrics are near zero	The oracle run still gives a clean tracker comparison; the detector failure becomes a domain-gap finding and the UVH-26 detector moves from stretch to core.

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